

Name: _____

Class: _____

Date: _____

Lab 4: Plan → Build → Test: Tip Calculator

Python Lab · Section 1.3 Program Design & Development



Goal

Practice investigating, designing, and building a program incrementally, then revising it iteratively from feedback.

Warm-up: Predict

```
bill = 40
tip = bill * 0.2
print(bill + tip)
```

Predict the output. What would you test next?

AP Connection

- Incremental: build and test one piece at a time.
- Iterative: revise based on feedback and testing.
- Document throughout with comments; credit any outside code.

Build It

Step 1. INVESTIGATE: Ask 2 classmates what they'd want in a tip calculator (split the bill? choose tip %?). Write 3 requirements as comments.

```
# Requirement 1: ...
# Requirement 2: ...
# Requirement 3: ...
```

Step 2. INCREMENT 1: Get the bill amount and print a 15% tip. Test it before moving on!

```
bill = float(input("Bill: $"))
tip = bill * 0.15
print("Tip:", round(tip, 2))
```

Step 3. INCREMENT 2: Let the user choose the tip percent. Test again.

```
percent = float(input("Tip %: "))
tip = bill * percent / 100
```

Step 4. INCREMENT 3: Split the total among a number of people.

```
people = int(input("How many people? "))
each = (bill + tip) / people
```

Step 5. ITERATE: Trade with a partner. Write down one piece of feedback you received and the change you made.

Date:

Build It (continued) · Level Up

Spicy What happens if `people = 0`? Test it, name the error type, and prevent it (preview of 3.6).

Write it in AP pseudocode.

Which steps today were incremental? Which were iterative?